

React JS



Week 10 :-

contents in react.js :-

React core syntax & JSX
working with components
working with Data

How to create React.js APP :-

1) install node.js

2) in the same folder install react App.

3) npx create-react-app sample

↳ project name means

4) npm start

a sample folder will be created

↳ for starting
react App:

Note: In sample folder we are having

folders

files

node_modules

public

src

package.json → very imp.
contains all dependencies
all node modules we
are using & their
versions

package-lock.json

README.md

contains App.js file which contains its content will be
displayed web browser through index.html file

↳ contains index.html contains browser contents like
title, etc.

ex: ReactApp

Note: defaultly the localhost:3000

↳ default

• Delete the code in App.js file & type `return` in

VS code & write `<h1>hello world</h1>`

will be displayed on

• change React App to sample App by
modifying index.html

browser.

lecture 11

components in reactjs : There are 2 types

- 1) functional component to get → - type raff in App.js file
- 2) class component to get → - type in vscode of rec in App.js file

difference b/w functional & class component :

render() & inheritance concept used

↓ → presented in class component but not in functional component
renders content to the browser.

Note : class components used rarely than functional

→ reactjs hooks can only be applicable to functional components : We are mostly using " .

react hook syntax

any variable to change name's content
↑ variable
const [name, setName] = useState("telugu skillhub")
↓ stored

lecture 12

state & props in class component in reactjs :

state : k/w used to initialize multiple variables with any value

props : k/w used to send variables/attributes from 1 file to other file in the same directory.

state syntax : display.js file need to create in the same

state = {
 name : "telugu skillhub"
 age : 12
}

props syntax

<h2> {this.props.name} </h2>

Note : see program for better understanding

Lecture - 13 ^{css}

How to use ^{↑ styling} styling in css Reactjs :-

Note:- Here for applying css styles we used functional component.

Note:- Here App.css file need to create to apply the css styles in 'App.js' file.

→ used for id
• → used for class } in css

Lecture - 14

onclick event handler in Reactjs :-

Syntax

```
<div>  
  <h1 onclick = { () => console.log("clicked")} > click here </h1>  
</div>
```

o/p :-
If click on the text

click here	clicked
------------	---------

console of browser (chrome)

Lecture - 15

How to use useState in Reactjs :-

↓
It is hook which allows us to track ~~state of any variables~~ state in a functional component. ^{or} arrays, strings etc. _(obj) objects

Note see program.

Note:- ex. to know how many time a button is clicked.

→ useState react hook used to create variables & to track them

Syntax import React, {useState} from 'react';

(const App = () => {

const [user, set-Value] = useState("Anil")

↑
initially "Anil" assigned to user if i want to update user variable by set-value.

syntax <form onSubmit={submitHandler}> user defined event
 <input />
 <input /> { some i/p forms like textbox, passwords, emails, ph.no.s } etc.
 </form>

lecture - 19

How to use map() function directly :-

used to hydrate/map an array of values into an array of virtual component / variable as parameters / values
 (get elements / on)

syntax array.map('value, index') => <li key={index}>
 [1, 2, 3, 4, 5] 0, 1, 2, 3, 4 2 value
 should be unique

note - simply map() takes an arrow func. as i/p parameter.

lecture - 20

How to use filter() :-

It is used to filter an array of values / array of object - based on some particular condition/s.

syntax :-

array.filter((variable) => condition...)

eg find the elements in the array which includes letter 'j'.

lecture - 21

export vs export default in reactjs :-

Both are used to export the content of a file to itself to another file.

export default :-

export default dashboard

import **Dashboard** from './dashboard';

Syntax import **&** component

export :-

export const dashboard = () => {

import 2 component }

lecture - 22 :-

How to create a login form in reactjs :-

It contains only 2 i/p's **username** & **password**
(or email)

lecture - 23 :-

How to create a signup form in reactjs :-

It contains more than 2 i/p fields like →
username
email
password
~~id = number~~
confirm - password

lecture - 24 Note :- (Continuation of form validation in reactjs :-

→ This concept is used to validate the i/p's given in the form like :-

- 1) username must contain min. of 4 letters.
- 2) password is equal to confirm password etc.

Based on these 2 conditions the form will be validated.

lecture - 25 :-

How to create a calculator App in reactjs :-

→ used to hold the expression

Result → if we click then we will get

result is :-

□ □ □ □ □

□ □ □ □ □

□ □ □ □ □

result of
the entered expression

Lecture - 26

How to Get API data using `Fetch()`

It is used to fetch/get the API data using the `url` (or) to make http request to get the data from API.

Notes `fetch()` is mostly used in useEffect react hook.

Notes use a fake url to get the API.

It takes an arrow func. & a dependency as parameters.

Notes Here we should convert the API data into json-format.

Lecture - 27

How to GET API data using `AXIOS`

→ `AXIOS` it is a js library to make any type of http request like

- get
- post
- delete
- update

} ex. for http requests

Notes for installation of `AXIOS` command

`npm install axios` (in src folder)

Notes `AXIOS()` is mostly used in

`useEffect` react hook. Ex: `axios.get('API url')`

Notes Here no need to convert the API data into json format

Syntax `useEffect(() =>`

`2 axios.get('http://...').then(res => setData(res.data))`, `[]`).

↳ dependency.

Lecture - 28

How to use `Firestore` real-time Database in `reactjs`

i.e. `Reactjs` form submission to firebase Database

* `firebase` is nosql database means there is no tables means all the data is stored in the form of JSON format.

application :-

- ⊗ Submitting reactjs form to the backend (or) Realtime time database.
- ⊗ see the process of installing the realtimedb.
- ⊗ finally a link/url will be generated for the realtime timebase db.

code :-

```
import axios from 'axios';
```

```
const submitHandler = e =>
```

```
2 e.preventDefault();
```

```
    axios.post('https://teluguskillskillhub.com/register',  
              data).then(() => alert('submitted successfully'));
```

firebase db url/link
form data stored in file
100
100

Note :- in the registered.json file our form data will be stored.

forms

full name
email
password
confirm password

Realtime firebase db :-

➡ http://teluguskillskillhub.com
teluguskillskillhub100
• register

→ if we click on register we can see the data in the register file

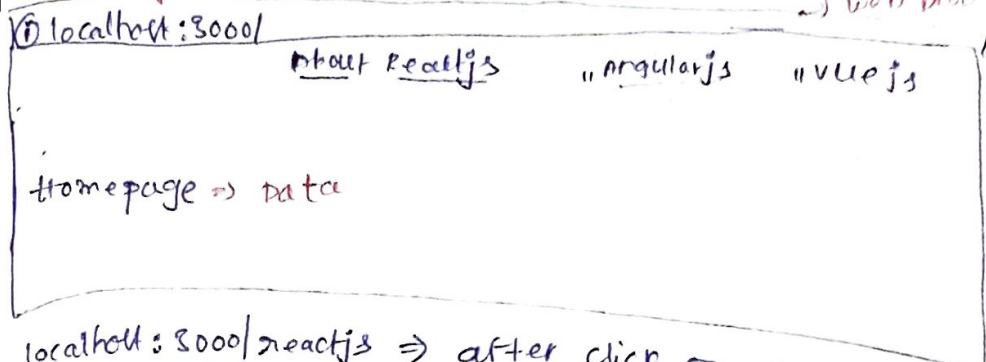
0/100

0/100	submitted successfully
submit	submit

Lecture - 29%

Concept 1. React Router

It is used to develop a multipage application.
multipage application :-

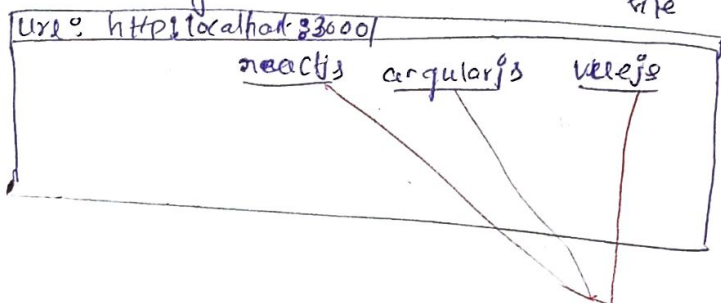


localhost:3000/reactjs => after click on About reactjs
the url will be changed like this.
"/angularjs" => About angularjs
"/vuejs" => About vuejs

Single page application :-

In this application we don't make http request multiple times

Here only one http request is made only for that request only the content in the web file is loaded



if we click on any link without changing the url the link's content will be displayed

Advantages :- all links content

only 1 link's content

multipage app.

- entire content / unwanted content will not be loaded once
- fast response than single page app.

single page app.

- it loads unwanted content along with wanted content
- response delay

To install React Router :

Command : `npm install react-router-dom`

It is a module
components of react-router-dom module :

⊗ import { BrowserRouter, Switch, Route }

These 3 components used for routing purpose

⊗ import { Link }

↳ used to redirect from 1 link to other required link by clicking a link on website.

Syntax :- if we want to include any component on browser routing it should be written this way.

`<BrowserRouter>`

`<Switch>` used for getting / change focus to exact path

`<Route path="/" exact component={*} />`
↓
component url path
↓
component Name

if we click on the any exact component then the website

will redirect to the particular given Route path.

Lecture 30 :-

Concept : Redirect component in reactjs.

It is used in developing multipage apps for Redirecting from one url to another url.

Syntax if (auth)

2 return `<Redirect to="/dashboard"/>`
}

Note : website will redirect from current page to dashboard page. using this `<Redirect>` tag.

lecture - 31 :-

concept :- useHistory hook in reactjs :-

This hook is used to redirect from one page/component to another page or component in the multipage application.

note :- The work of useHistory hook & Redirect component are same. but difference is we can use useHistory hook within functional level components. but we can't use in class level components.

note :- Redirect component we can use both either in functional (or) class level components.

lecture - 32 :-

concept :- How to use URL parameters in reactjs :-

⊗ path params Vs Query params

note :- These 2 params widely used in developing multipage application. i.e. if we want to send a parameter from 1 page/component to another page it can be done with help of http request url along with parameter
(or)
http request in url.

⇒ Now we can send the parameters in 2 ways :-

1) path params

2) Query params

↓
use if we know the variable of value (or) parameter name

↓
Here no need to know about the variable name

path params :-

<Route path = "/profile/:name" component = <profile />

https://localhost:3000/profile/telugustillhub

Variable name = telugustillhub as assigned

Query params :

both are matched.

< Route path = "/profile" Component = { profile } />

http://localhost:3000/profile?name=teluguskills
Variable $\Rightarrow$ name = teluguskills
param name Variable param Value

Note : in Query params "?" will be used.

Lecture - 33 :

Implementing path params

(actually he did ~~not~~ implement Query param concept)
in lecture - 32
getSelf

Lecture - 34 :

Concept : how to import images in reactjs :-

code : import ReactLogo from 'react.png' ;
in same directory

return (

<h1> React logo </h1>

Lecture - 35 : how to import audio in reactjs :-

code : import AudioFile from 'music.mp3' ;

return (

<audio controls> $\rightarrow$ forgetting stop & play & down options
& song skipping options

<source src={AudioFile} type="audio/ogg" />

</audio>

)

note : we should curly braces when we are using
javascript code with in the html code.

lecture - 36 - How to import a video file in reactjs.

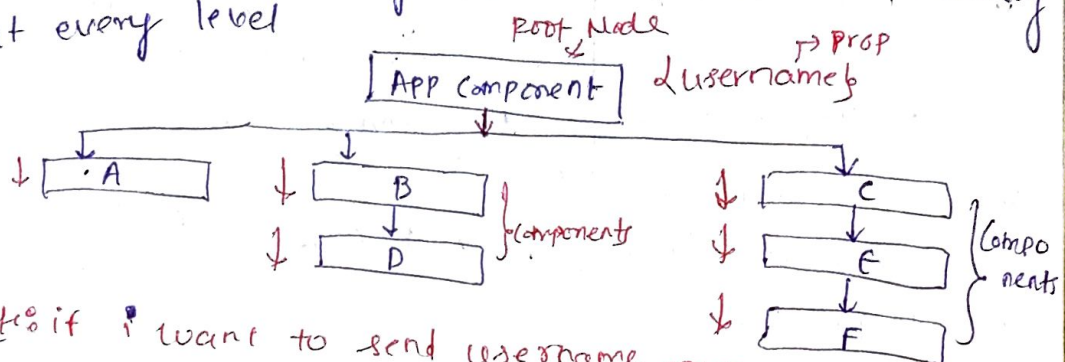
code :-

```
import VideoFile from './video.mp4';  
return (  
  <video width="320" height="250" controlling <stop> <start> <download etc.>  
    <source src={videoFile} type="video/mp4" />  
  </video>  
);
```

↓
Video file type

lecture - 37 - How to use useContext hook in reactjs.

It is used to pass data through the components tree without having to pass props down manually at every level.



Note :- if I want to send username prop from Root Node to F component I need send username prop for components C, E (here for C, E components).

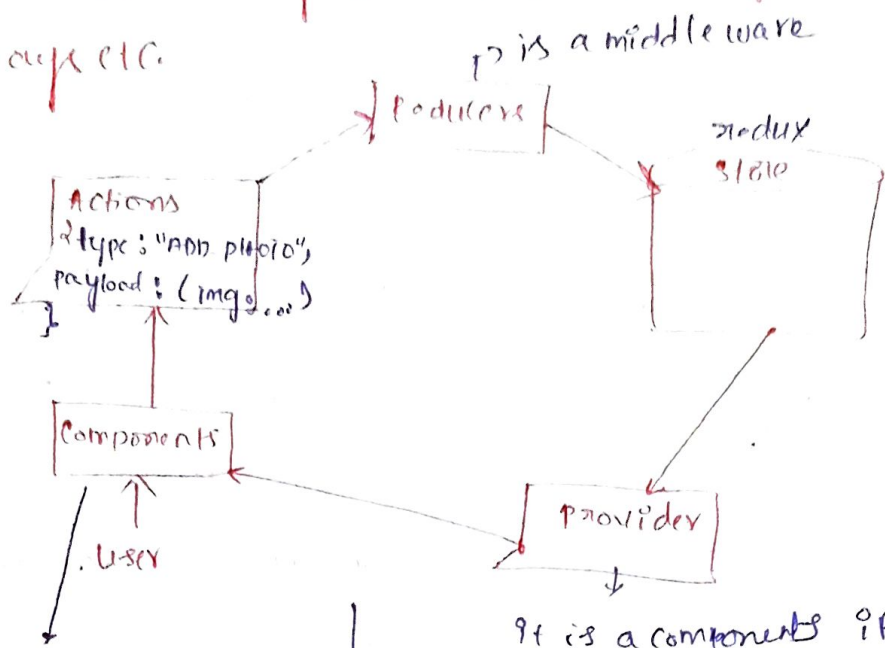
The username prop is not required but by using useContext hook concept we can directly send the props to any component in the tree.

lecture - 38 - Redux in reactjs :-

Redux concept is used to declare state variable in global variable.

→ Here ^{it} is all about managing state value/variables in multiple components at a time, using "redux state" ^{is like a box}.

① Redux state: may store variables, integers, objects, arrays etc.



In to-do application
if I want add
(or) remove a task
from/to the application

in that case we need to
click submit button in that case an action will
be triggered

type: type of action i.e. to do or completed action
payload: data of the action.

Reducers: - it checks the type of the action
& updates application state with the payload in the state.
{ add to do actions & removes completed actions }

Dependencies for Redux Concept:

- 1) npm install redux
- 2) npm install react-redux

some times we will use data in redux, i.e. react
(or) we will performing some operations by combining
react & redux in that case module will be used.

3) npm install redux-thunk :-

It is a middleware it handles function dispatching.

Ex:-

Count from App Component: 0 ^{if we increment} + 2
Count from Inc Component: 0 + 2

Increment

Decrement

if we click on increment button then the global variable count in App, Inc components will incremented. Similarly if we click on decrement the count value will be decremented.

Note:- Compare this Ex: with this definition of Redux - store / Redux

Redux is nothing but managing state values in multiple components at a time using Redux store.

Lecture - 39:-

mapStateToProps concept in reactjs :-

This concept used to :-

The data stored in the react store, to use the data in react components.

(OR)

It is a function that you would use to provide the state data to your component.

Ex: A/c to above provided example "Count" is a variable which is stored in react-redux store. Now here we seeing how we are using

the react ~~data~~ medium store variable in the components (App & Inc).

Note: it see lecture 40 notes is regarding props concept.

lecture - 41 :-

concept :- Life cycle methods in reactjs :-

① life cycle refers to the process of executing methods

② generally lifecycle or ~~methods~~ are same side effects.

Note: generally we will use hooks (react) in functional components only

react hook
↑

side-effects in functional components : use Effect()

class based components can't use react hooks

class based component life cycle :-

some of life cycle methods or side effects are :-

componentDidMount() :- It can be used when we

want to perform an action after immediate completion of return statement.

Note: in functional components use Effect() hook is also works after immediate execution of return statement

DOM -> document object model => it refers to the

data displayed on browser.

→ its syntax similar as use Effect(...., [])

componentDidUpdate() :- it can be used to update or add values to our react redux store / react project.

Ex: adding an item to cart in flipkart
it's syntax similar as `useEffect(..., [someValue])`

componentWillUnmount() :- it is used to remove an item / data value

Ex: removing an item from cart of flipkart account

It's syntax similar as $\Rightarrow$

`useEffect(()  $\Rightarrow$  return ()  $\Rightarrow$  { ... }, [])`

Now see the corresponding program for better understanding :-

Show :- This variable used for part-1 / above part for deleting & showing the part-1

Count :- variable used for end part / below part, used for updation purpose.

o/p $\Rightarrow$ hellow world! part-1 / above

delete Header $\rightarrow$ if we click heading dis appeared

Count : 0

part-2 / below

Increment $\rightarrow$ if we click count incremented.

Lecture - 42 : concept : Bootstrap with reactjs ^{collection predefined html classes}

- means how to use bootstrap within our reactjs
- Bootstrap is a good user interface framework, to create a best front end design.
- for using bootstrap in reactjs we need to install module $\Rightarrow$ react-bootstrap

Note if we want use css in react-bootstrap for that purpose we need to copy & paste the link tag ~~to~~ from official website "react-bootstrap.github.io" into the index.css file

```
import { Jumbotron, Button } from 'react-bootstrap';
```

export $\uparrow$

```
import Jumbotron from 'react-bootstrap';  
import Button from 'react-bootstrap';
```

$\Rightarrow$ default export.

Max. no. of people will use export approach for importing the components/modules in reactjs. otherwise we need to ~~install~~ import all components individually.

Jumbotron : It is a component of bootstrap, which is used to showcase / display the important / key content on our website.

steps:

- 1) Install react-bootstrap
- 2) copy link tag
- 3) Import components

Lecture - 43 : concept : how to use material ui in reactjs

- material-ui is a front-end framework, if use this then application will be high responsive

→ material-ui official website: material-ui.com

installation: `npm install material-ui`

How to use material ui Components in react code?

Now click on Components select any component on menu to ex select Tab component

material ui

React - bootstrap

→ Doesn't provide info. about → It will provide →
How to import Components

Note: The same Component output will be displayed on react. o/p ui after copying the code of tab Component.

Note: Lecture 44 is a double session

Lecture - 45: concept: how to download any files in reactjs. img/audio/video/.txt etc

href ⇒ hyper text reference

↳ give the filename to the href - otherwise the downloaded file will be saved with a random name.

Note: but the file which we want to download which is to be in the current local directory.

Lecture - 46: concept: send emails in reactjs -
using the emailjs module
or it is an email service provider

→ This module mostly used to submit the contact forms

official website for emailjs

Installation of emailjs website - npm install emailjs-com

→ Now there there are 3 variables which will change from 1 user to another user.

we should type

→ To create an email account there are 2 impo steps:

1) email services

2) email templates ⇒ format of received mail

→ username will be on the API keys (init())
+ variable of this variable

Note: in Javascript if we want to use curly braces ⇒ CSS style tags
style = {} : write code of CSS

Lecture 47: Concept: Search filter app in reactjs

filter(): used filter some value from the data.

syntax: data.filter((city) ⇒ city)

Ex: project: searching a district name out of all district names.

Lecture 48: Concept: Reactjs Interview Q & Answer

Lecture - 49: Concept: React redux tutorial (or)

the implementation of redux concept

for this implementation we need to install few dependencies :-

npm install redux

npm install react-redux

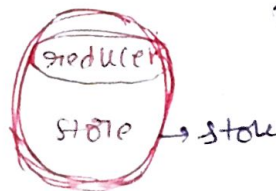
npm install redux-thunk type

npm install redux-devtools-extension

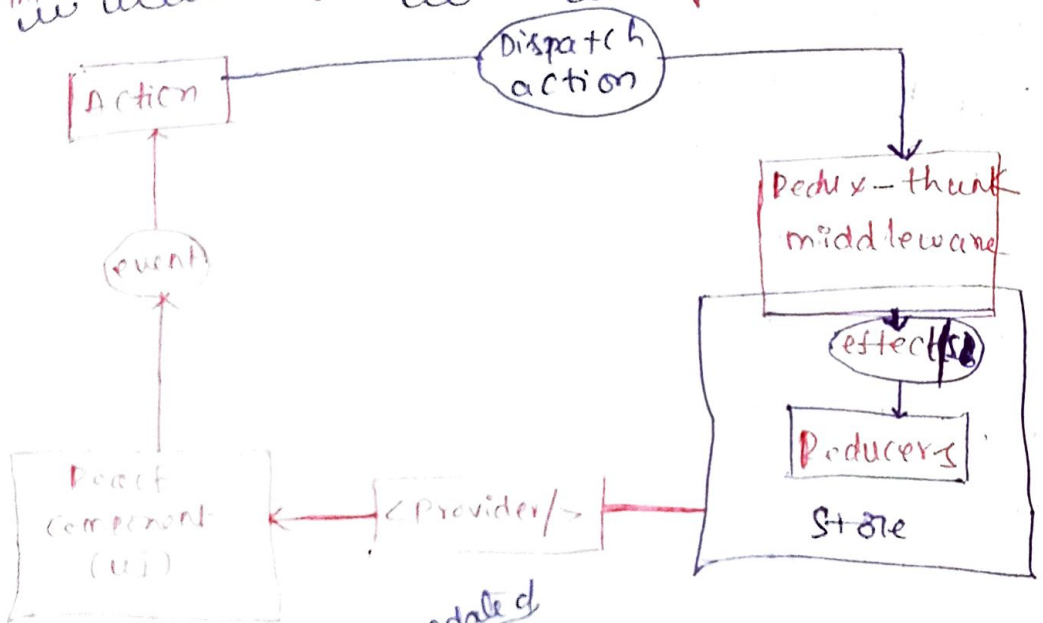
major terms in Redux

Actions

means within the store the reducer is there.



Implementation of redux life cycle :-



Reducers returns an ^{updated state} variable/data value to the store based on action dispatched.

→ Redux-thunk is a middleware, it is used to handle

dispatch the actions/functions properly.

* Store is used to store the state variable/values

* Reducer is a part of Store

→ if u want to use state values (variables),
the react component (App.js) use <Provider>
component.

Note: provider is a module of react - redux module
used to create a store of react

* `createStore(reducer, state)` ; parameters

* `reducer(state, action)`

* `action { type, payload }`

↓
Name of the variable
which we want to change ↓
amount of data

Steps in implementation of redux Concept :-

- 1) 1st install packages
- 2) create react store 1st → i.e. `store.js` file
- 3) create reducer 2nd → i.e. `reducer.js` file
- 4) Don't overwrite the `index.js` file just remain it as original.

→ add, redux-devtool extension in chrome

Note: with the help of `mapStateToProps` we can use
all the values in the store in the react
Component i.e. App.js file.

`local-variable ; state = 5` → is displayed in

redux devtool extension
when we add it to
browser.

→ we are using state values in the react Component
(App.js) using connect module

- 5) create action.js file. , it is used to update
values in store

⊗ execution of redux tracing :-

apps → action.js → reducer.js → Store.js

↓
action type, payload declared with some values

↓
depends on action type
action will be performed & returns values (updates)

lecture - 50% Concept :- class Component concept
leave it if not so much imp.

lecture - 51% Concept :- How to use useRef hook in reactjs :-

~~lecture - 52% Concept :-~~ This hook is used to take the value from the user and assign it into a variable. content

Note :- prop-render means payload

This hook used to :- ① auto focusing the '911' field
② we can store our previous values using useRef hook.

→ Note :- by using this hook we can assign user values to the variable without using any event handler.

ex :- onChange not required instead which we can use ref attribute

ref = &data user typed data stored in the data variable.

Note :- for auto focusing purpose, this hook uses the useEffect hook. becaz it is executed after return statement is executed.

lecture - 52 : concept of use Reducer() tool

This hook() work flow is same as redux

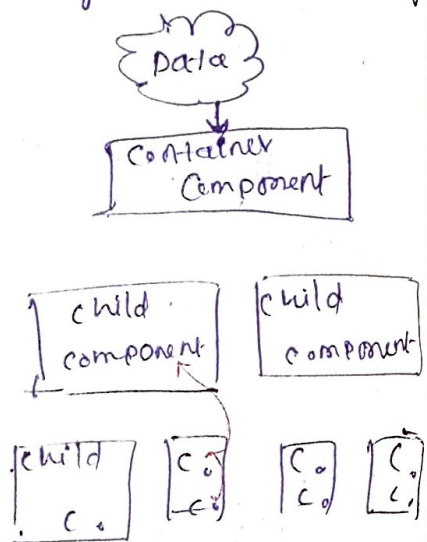
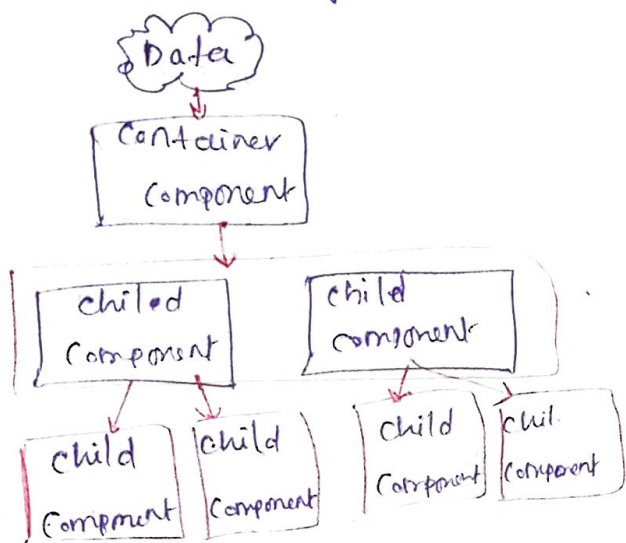
Note: reducer will return the modified state value to the store based on the user action.

Dispatched:

Note: state value $\Rightarrow$ ex: $\text{const } [a, \text{xet-a}] = \text{useState}()$
 $\downarrow$
 is a stat value

Lecture - 53% - Concepts Context API Concept

→ Context API is a kind of ^{extra} feature used to share data with multiple components / multiple files (.jsx) without passing the data through props manually



called context API

prop drilling

I means using props concept.

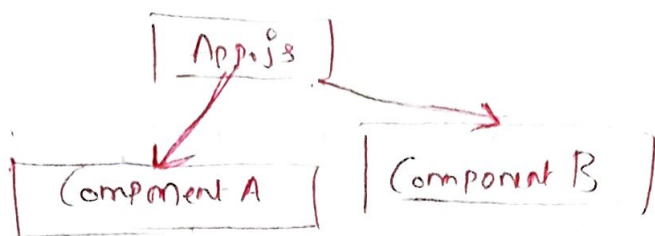
Here Data sent to all intermediate components even if the Data is not required

Micro data sent to
required a child component
directly without
wasting of time &
complexity

Wasting of time & Complexity

Exo

(state - values)



lecture - 54% Concept : Higher order Component in Reactjs. (HOC)

HOC : It is used for reusing the Component logic

Ex: if u want to move / redirect from 1 page (website) to another page we require the user authorization. Basically it is logic used for login & sign up page.
Ex: from Home page to dashboard page.

Notes

HOC(App)

→ App component will be wrapped with HOC

Component -

→ read more about HOC in next page →

lecture - 55% Concept : useMemo hook in Reactjs :-

This hook is used to improve the entire render ~~factor~~ performance of the App. when we performing most expensive function.

Notes - generally useEffect(), useMemo hooks execute after, during the rendering of entire code accessing (or) executing.

Notes: useMemo will not run for every re-render happens.

→ It will run during the 1st render and when its dependency values change.

✓ useEffect() will run after render of the Component (on access/execution)

⊛ useMemo() will run during render of the Component (on execution)

Note: if we use useState() then the given Component

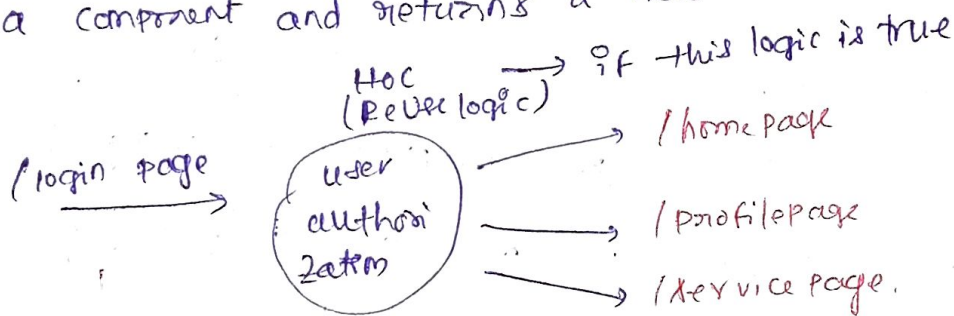
is recalculated on re-rendering means entire code is again executed unwantedly. ∴ The performance is degrading.

Syntax: useEffect $(() \Rightarrow \{ \}, [a, \dots])$
if this array modified then the hooks called useMemo $(() \Rightarrow \{ \}, [a, \dots])$
↓
Dependency array

Lecture - 54 Continuation of

→ HOC is a pure func. with zero side effects & doesn't modify the i/p Component

→ HOC is a ~~Component~~ is a function that takes a Component and returns a new Component.



Lecture - 56: Concept: useEffect vs useMemo have already covered this topic in lecture 55

Note: useContext() is used to access/use the values which are represented in the react store.
~~Delete the items from~~

Lecture - 57 :- Concept :- Stateful (VS) Stateless Components.

Stateful Component :- if we declare the state values using the useState hook & handle them in any component instant call it is called Stateful Component.

Stateless Component :- It is quite opposite to the Stateful Component. i.e. even in this component the state variables are not declared using the useState hook. i.e. only if it contains the HTML contents or it with using props.

Lecture - 58 :- Concept :- Controlled vs Uncontrolled Components :-
→ tag

Controlled Component :-

The component in which we can send the change handler, variable or props to the text component file.

Uncontrolled Component :- This component does not need

to contain the callback func's. Instead of using the callback func's it uses as follow

→ used for getting the data from user
Input ref = { } / >

Lecture - 59% Concept: React lazy loading

bundling: Converting multiple ^{js} files into a single ^{on} any other file
a single .js
on any other files

Exo login, home, dashboard.js files are bundled into single file called App.js in reactjs.

React.lazy() is used for Dynamic importing of other components / files into the current file (Exo App.js file).

→ In Normal importing all imported components are loaded into the current App.js file at once without accessing any one required component. file is some what time taken.

→ But in Dynamic importing if we import all components / files into App.js file (current file) Dynamically. Then all components are not get loaded at once, the dynamically imported components loaded individually also to request made for other component is accessing of a required file is less time taken. 1. App's efficiency is somewhat increasing.

↑
creating error component as
undefined

Lecture - 60% Concept: <ErrorBoundary /> in reactjs :-

→ <ErrorBoundary /> used to handle the errors in the react project

Error boundary Component will catch javascript errors anywhere in their child Component tree and Display corresponding error of fallback ui

Ex: `<ErrorBoundary>` ← components used inside this tag
`<Component A1>` → child Component
`</ErrorBoundary>`

`<ErrorBoundary>` is not a react inbuilt importing Component but it is a custom component i.e. it is developed by the user. ↳ user defined

fallback UI → UI of error i.e. ex: ~~oops~~ something went wrong

→ ComponentDidCatch() this method { ~~page~~ page not found, etc.

will get works whenever an error occurs.

In the example code we implemented 6,

⊙ after incrementing 5 we will get an error in the react app. i.e. ^{due to} Component A

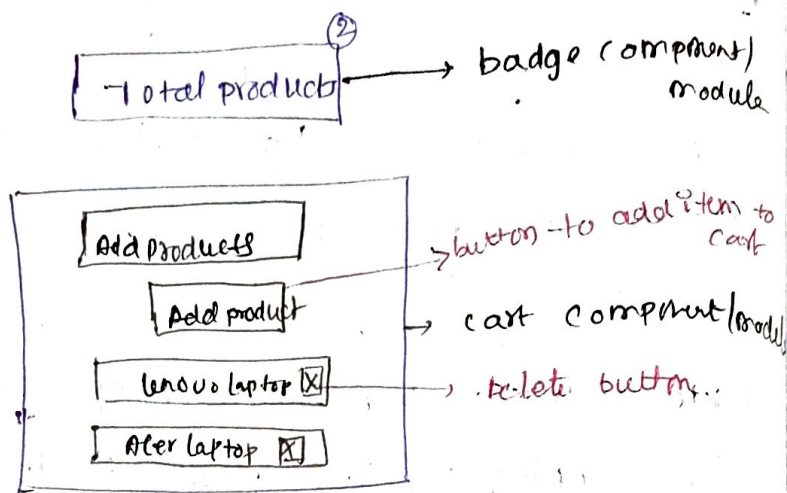
→ but Component B don't have any error

⇒ app crash or occurrence of error in the app.

How to establish connection b/w Node.js
(backend or server) & React.js (front-end or client) :-

- 1) Start the ~~node~~ writing the code for both server & client
- 2) Now start the backend server using the command $\rightarrow$ node filename.js
Ex: node index.js
- 3) Now take port number of server, which is running on the port number and assign the port number to the client in the code
- 4) Now start the client server using command :- npm start
- 5) Now send the data from server it will displayed on the client server
- 6) $\therefore$ connection b/w server & client is established successfully.

Understanding react - redux concept with Shopping
cart app - (for understanding this concept better)
we use react-devtools extension
in chrome



import {connect} from 'react-redux';

↓
This component/module used to access the
react store in any react components in our react
project.

→ This connect component consist of 2 things
1) mapStateToProps
2) mapDispatchToProps

↓
by using this we can
access the values/data
in our react store.
we can store the accessed
value in a local variable.

ex: used to know the no. of
products in the cart

email, password auth.
React firebase authentication :- using firebase

by using this concept the user can access the dashboard (of app) by providing the same credentials (pswd, username, email etc.) to signup, signin/login.

if ~~both~~ ^{we are working with login form} } for creating an account for accessing the homepage (or) dashboard
→ in the firebase project creation we have to steps
1) authentication ⇒ enabling email
2) configura. ⇒ $\langle / \rangle$ ⇒ web project (project type)
after that a code will be generated use the code in our project by creating file `firebase.js`
Deploying react project in firebase :-

for that we require a firebase account & firebase project. & by typing some firebase deployment commands in terminal we can deploy our project into firebase then finally a link (http) will be generated in terminal by using that link we can access the o/p of our react project in chrome browser.

→ i.e. the generated link is allowed by google & google trust ~~the~~ our react project ~~as~~ as a secure connection becaz we deployed our project in the firebase.

∴ In websites we can create signin with google button with the help of react and firebase (integration).

Installation of firebase :- `> npm install firebase`.

Notes: 1st ~~not~~ ⁴ ~~uploaded~~ ^{deployed} weather app project to firebase project

Note: for signin into the firebase db & signout
from the firebase db some inbuilt functions will
be used in our project see the
firebase authentication in reactjs project.

UI

Authentication

enter email

enter password

signin

signup

login

Note: useEffect()

- this react hook is used for side effect i.e. this hook is executed after entire code in rendering / executed if there is no dependency parameter
- if one user is login then the logged user details will be ^{for signin} displayed by using `useEffect()` react hook.
after clicking signin / signup button.